

Module 1 Lesson 3

The Rational Numbers (5.3)

Objectives

In this lesson, we will learn how to

- define the rational numbers;
- reduce rational numbers;
- convert between mixed numbers and improper fractions;
- express rational numbers as decimals;
- express decimals in the form $\frac{a}{b}$;
- multiply and divide rational numbers;
- add and subtract rational numbers;
- use the order of operations with rational numbers;
- apply the density property of rational numbers;
- solve problems involving rational numbers.

1 Defining the Rational Numbers

Rational Numbers

The set of rational numbers is the set of all numbers that can be expressed in the form

$$\frac{a}{b},$$

where a and b are integers and $b \neq 0$.

- The integer a is called the numerator.
- The integer b is called the denominator.

Examples of rational numbers include

$$\frac{1}{4}, \quad -\frac{1}{2}, \quad \frac{3}{4}, \quad 5, \quad 0.$$

Notice that integers are rational numbers because they can be written with denominator 1:

$$5 = \frac{5}{1}, \quad 0 = \frac{0}{1}.$$

2 The Fundamental Principle of Rational Numbers

Fundamental Principle of Rational Numbers

If $\frac{a}{b}$ is a rational number and c is any number other than 0, then

$$\frac{a \cdot c}{b \cdot c} = \frac{a}{b}.$$

The rational numbers

$$\frac{a}{b} \quad \text{and} \quad \frac{a \cdot c}{b \cdot c}$$

are called equivalent fractions.

Reducing to Lowest Terms

To reduce a rational number to its lowest terms, divide both the numerator and denominator by their greatest common divisor.

Example 1: Reducing a Rational Number: Reduce

$$\frac{130}{455}$$

to lowest terms.

Solution

Begin by finding the greatest common divisor of 130 and 455.

$$130 = 2 \cdot 5 \cdot 13$$

$$455 = 5 \cdot 7 \cdot 13$$

The common factors are 5 and 13, so the greatest common divisor is

$$5 \cdot 13 = 65.$$

Now divide both the numerator and denominator by 65:

$$\frac{130}{455} = \frac{130 \div 65}{455 \div 65} = \frac{2}{7}.$$

Thus,

$$\boxed{\frac{130}{455} = \frac{2}{7}}.$$

3 Mixed Numbers and Improper Fractions

Mixed Number

A mixed number consists of the sum of an integer and a rational number, expressed without the use of an addition sign.

For example,

$$3\frac{4}{5}$$

means

$$3 + \frac{4}{5}.$$

Improper Fraction

An improper fraction is a rational number whose numerator is greater than its denominator.

For example,

$$\frac{19}{5}$$

is an improper fraction because 19 is larger than 5.

4 Converting a Positive Mixed Number to an Improper Fraction

Steps

To convert a positive mixed number to an improper fraction:

1. Multiply the denominator of the rational number by the integer and add the numerator to this product.
2. Place the sum over the denominator of the mixed number.

Example 2: Converting a Positive Mixed Number to an Improper Fraction:
Convert

$$3\frac{4}{5}$$

to an improper fraction.

Solution

Multiply the denominator by the whole number, then add the numerator:

$$5 \cdot 3 + 4 = 15 + 4 = 19.$$

Place this result over the original denominator:

$$3\frac{4}{5} = \frac{5 \cdot 3 + 4}{5} = \frac{19}{5}.$$

Thus,

$$\boxed{3\frac{4}{5} = \frac{19}{5}}.$$

5 Converting a Positive Improper Fraction to a Mixed Number

Steps

To convert a positive improper fraction to a mixed number:

1. Divide the denominator into the numerator.
2. Record the quotient and the remainder.
3. Write the mixed number as

$$\text{quotient} \frac{\text{remainder}}{\text{original denominator}}.$$

Example 3: Converting from an Improper Fraction to a Mixed Number:
Convert

$$\frac{42}{5}$$

to a mixed number.

Solution

Divide 42 by 5:

$$42 \div 5 = 8 \text{ remainder } 2.$$

So the quotient is 8 and the remainder is 2.

Therefore,

$$\frac{42}{5} = 8\frac{2}{5}.$$

Thus,

$$\boxed{\frac{42}{5} = 8\frac{2}{5}}.$$

6 Rational Numbers and Decimals

Writing Rational Numbers as Decimals

Any rational number can be expressed as a decimal by dividing the denominator into the numerator.

Example 4: Expressing Rational Numbers as Decimals: Express each rational number as a decimal.

a. $\frac{5}{8}$

b. $\frac{7}{11}$

Solution

a.

$$\frac{5}{8} = 5 \div 8 = 0.625.$$

So,

$$\boxed{\frac{5}{8} = 0.625}.$$

This decimal stops, so it is called a terminating decimal.

b.

$$\frac{7}{11} = 7 \div 11 = 0.636363 \dots$$

So,

$$\boxed{\frac{7}{11} = 0.\overline{63}}.$$

The digits 63 repeat indefinitely, so this is called a repeating decimal.

7 Expressing Decimals as a Quotient of Two Integers

Terminating Decimals

Terminating decimals can be expressed with denominators of

10, 100, 1000, 10,000, and so on.

The digits to the right of the decimal point become the numerator of the rational number.

Example 5: Expressing Terminating Decimals in a/b Form: Express each terminating decimal as a quotient of integers.

a. 0.7

b. 0.49

c. 0.048

Solution

a.

$$0.7 = \frac{7}{10}$$

because the 7 is in the tenths position.

$$0.7 = \frac{7}{10}$$

b.

$$0.49 = \frac{49}{100}$$

because the 9 is in the hundredths position.

$$0.49 = \frac{49}{100}$$

c.

$$0.048 = \frac{48}{1000}$$

because the 8 is in the thousandths position.

Now reduce:

$$\frac{48}{1000} = \frac{48 \div 8}{1000 \div 8} = \frac{6}{125}.$$

Thus,

$$0.048 = \frac{6}{125}.$$

8 Expressing Repeating Decimals in Fraction Form

Example 6: Expressing a Repeating Decimal in a/b Form: Express

$$0.\overline{6}$$

as a quotient of integers.

Solution

Let

$$n = 0.\overline{6} = 0.6666 \dots$$

Since there is one repeating digit, multiply both sides by 10:

$$10n = 6.6666 \dots$$

Now subtract the first equation from the second equation:

$$\begin{array}{r} 10n = 6.6666 \dots \\ -n = -0.6666 \dots \\ \hline 9n = 6 \end{array}$$

Solve for n :

$$9n = 6$$

$$n = \frac{6}{9}$$

Reduce:

$$n = \frac{2}{3}.$$

Therefore,

$$\boxed{0.\overline{6} = \frac{2}{3}}.$$

9 Multiplying Rational Numbers

Multiplying Rational Numbers

The product of two rational numbers is the product of their numerators divided by the product of their denominators.

If $\frac{a}{b}$ and $\frac{c}{d}$ are rational numbers, then

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{a \cdot c}{b \cdot d}.$$

Example 7: Multiplying Rational Numbers: Multiply. If possible, reduce the product to lowest terms.

$$\left(-\frac{2}{3}\right) \left(-\frac{9}{4}\right)$$

Solution

Multiply across:

$$\left(-\frac{2}{3}\right)\left(-\frac{9}{4}\right) = \frac{(-2)(-9)}{3 \cdot 4}.$$

Since a negative times a negative is positive,

$$\frac{(-2)(-9)}{3 \cdot 4} = \frac{18}{12}.$$

Reduce:

$$\frac{18}{12} = \frac{3 \cdot 6}{2 \cdot 6} = \frac{3}{2}.$$

As a mixed number,

$$\frac{3}{2} = 1\frac{1}{2}.$$

Thus,

$$\boxed{\left(-\frac{2}{3}\right)\left(-\frac{9}{4}\right) = \frac{3}{2} = 1\frac{1}{2}}.$$

10 Dividing Rational Numbers

Dividing Rational Numbers

The quotient of two rational numbers is the product of the first number and the reciprocal of the second number.

If $\frac{a}{b}$ and $\frac{c}{d}$ are rational numbers, and $\frac{c}{d} \neq 0$, then

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c} = \frac{a \cdot d}{b \cdot c}.$$

Example 8: Dividing Rational Numbers: Divide. If possible, reduce the quotient to lowest terms.

$$-\frac{3}{5} \div \frac{7}{11}$$

Solution

Change division to multiplication by using the reciprocal:

$$-\frac{3}{5} \div \frac{7}{11} = -\frac{3}{5} \cdot \frac{11}{7}.$$

Multiply across:

$$-\frac{3}{5} \cdot \frac{11}{7} = -\frac{3 \cdot 11}{5 \cdot 7} = -\frac{33}{35}.$$

Thus,

$$\boxed{-\frac{3}{5} \div \frac{7}{11} = -\frac{33}{35}}.$$

11 Adding and Subtracting Rational Numbers with Identical Denominators

Same Denominator Rule

The sum or difference of two rational numbers with identical denominators is the sum or difference of their numerators over the common denominator.

If $\frac{a}{b}$ and $\frac{c}{b}$ are rational numbers, then

$$\frac{a}{b} + \frac{c}{b} = \frac{a + c}{b}$$

and

$$\frac{a}{b} - \frac{c}{b} = \frac{a - c}{b}.$$

Example 9: Adding and Subtracting Rational Numbers with Identical Denominators: Perform the indicated operations.

a. $\frac{3}{7} + \frac{2}{7}$ b. $\frac{11}{12} - \frac{5}{12}$ c. $-5\frac{1}{4} - \left(-2\frac{3}{4}\right)$

Solution

a.

$$\frac{3}{7} + \frac{2}{7} = \frac{3+2}{7} = \frac{5}{7}.$$

$$\boxed{\frac{3}{7} + \frac{2}{7} = \frac{5}{7}}$$

b.

$$\frac{11}{12} - \frac{5}{12} = \frac{11-5}{12} = \frac{6}{12} = \frac{1}{2}.$$

$$\boxed{\frac{11}{12} - \frac{5}{12} = \frac{1}{2}}$$

c. First convert the mixed numbers to improper fractions:

$$-5\frac{1}{4} = -\frac{21}{4}$$

and

$$-2\frac{3}{4} = -\frac{11}{4}.$$

Then subtract:

$$-5\frac{1}{4} - \left(-2\frac{3}{4}\right) = -\frac{21}{4} - \left(-\frac{11}{4}\right).$$

Subtracting a negative becomes addition:

$$-\frac{21}{4} + \frac{11}{4} = \frac{-21+11}{4} = -\frac{10}{4}.$$

Reduce:

$$-\frac{10}{4} = -\frac{5}{2}.$$

As a mixed number,

$$-\frac{5}{2} = -2\frac{1}{2}.$$

Thus,

$$\boxed{-5\frac{1}{4} - \left(-2\frac{3}{4}\right) = -\frac{5}{2} = -2\frac{1}{2}}.$$

12 Adding and Subtracting Rational Numbers with Unlike Denominators

Unlike Denominators

If rational numbers have different denominators, we use the least common multiple of their denominators to rewrite the rational numbers. The least common multiple of the denominators is called the least common denominator, or LCD.

Example 10: Adding Rational Numbers with Unlike Denominators:
Find the sum:

$$\frac{3}{4} + \frac{1}{6}.$$

Solution

Find the least common multiple of 4 and 6:

$$\text{LCM}(4, 6) = 12.$$

So the LCD is 12.

Rewrite each fraction with denominator 12:

$$\frac{3}{4} = \frac{3}{4} \cdot \frac{3}{3} = \frac{9}{12}$$

and

$$\frac{1}{6} = \frac{1}{6} \cdot \frac{2}{2} = \frac{2}{12}.$$

Now add:

$$\frac{3}{4} + \frac{1}{6} = \frac{9}{12} + \frac{2}{12} = \frac{11}{12}.$$

Thus,

$$\boxed{\frac{3}{4} + \frac{1}{6} = \frac{11}{12}}.$$

13 Using the Order of Operations with Rational Numbers

Order of Operations Reminder

When simplifying expressions with rational numbers, use the order of operations:

1. Parentheses or grouping symbols
2. Exponents
3. Multiplication and division from left to right
4. Addition and subtraction from left to right

Example 11: Using the Order of Operations: Simplify:

$$\left(\frac{1}{2}\right)^3 - \left(\frac{1}{2} - \frac{3}{4}\right)^2 (-4).$$

Solution

Because grouping symbols appear, perform the operation inside the parentheses first:

$$\frac{1}{2} - \frac{3}{4} = \frac{2}{4} - \frac{3}{4} = -\frac{1}{4}.$$

Now substitute:

$$\left(\frac{1}{2}\right)^3 - \left(-\frac{1}{4}\right)^2(-4).$$

Evaluate the powers:

$$\left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

and

$$\left(-\frac{1}{4}\right)^2 = \frac{1}{16}.$$

So,

$$\frac{1}{8} - \frac{1}{16}(-4).$$

Multiply:

$$\frac{1}{16}(-4) = -\frac{4}{16} = -\frac{1}{4}.$$

Therefore,

$$\frac{1}{8} - \left(-\frac{1}{4}\right) = \frac{1}{8} + \frac{1}{4}.$$

Rewrite with common denominator 8:

$$\frac{1}{8} + \frac{2}{8} = \frac{3}{8}.$$

Thus,

$$\boxed{\left(\frac{1}{2}\right)^3 - \left(\frac{1}{2} - \frac{3}{4}\right)^2(-4) = \frac{3}{8}}.$$

14 Density of Rational Numbers

Density Property of Rational Numbers

If r and t represent rational numbers, with $r < t$, then there is a rational number s such that s is between r and t :

$$r < s < t.$$

This means that between any two rational numbers, there is another rational number.

Example 12: Illustrating the Density Property: Find a rational number halfway between

$$\frac{1}{2} \quad \text{and} \quad \frac{3}{4}.$$

Solution

To find the number halfway between two numbers, find their average.
First add:

$$\frac{1}{2} + \frac{3}{4} = \frac{2}{4} + \frac{3}{4} = \frac{5}{4}.$$

Then divide this sum by 2:

$$\frac{5}{4} \div 2 = \frac{5}{4} \cdot \frac{1}{2} = \frac{5}{8}.$$

Thus, the number halfway between $\frac{1}{2}$ and $\frac{3}{4}$ is

$$\boxed{\frac{5}{8}}.$$

Check:

$$\frac{1}{2} < \frac{5}{8} < \frac{3}{4}.$$

15 Problem Solving with Rational Numbers

Adjusting a Recipe

A common application of rational numbers involves preparing food for a different number of servings than what the recipe gives.

The amount of each ingredient can be found as follows:

$$\text{amount needed} = \frac{\text{desired serving size}}{\text{recipe serving size}} \cdot \text{ingredient amount in the recipe}.$$

Example 13: Adjusting the Size of a Recipe: A chocolate-chip recipe for five dozen cookies requires $\frac{3}{4}$ cup of sugar. If you want to make eight dozen cookies, how much sugar is needed?

Solution

Use the formula:

$$\text{amount needed} = \frac{\text{desired serving size}}{\text{recipe serving size}} \cdot \text{ingredient amount in the recipe.}$$

Here,

$$\text{desired serving size} = 8 \text{ dozen,}$$

$$\text{recipe serving size} = 5 \text{ dozen,}$$

and

$$\text{ingredient amount} = \frac{3}{4} \text{ cup.}$$

So,

$$\text{amount of sugar needed} = \frac{8}{5} \cdot \frac{3}{4}.$$

Multiply:

$$\frac{8}{5} \cdot \frac{3}{4} = \frac{8 \cdot 3}{5 \cdot 4} = \frac{24}{20}.$$

Reduce:

$$\frac{24}{20} = \frac{6}{5}.$$

As a mixed number,

$$\frac{6}{5} = 1\frac{1}{5}.$$

Thus,

$1\frac{1}{5}$ cups of sugar are needed.
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Summary

Key Ideas

- Rational numbers can be written in the form $\frac{a}{b}$, where a and b are integers and $b \neq 0$.
- Equivalent fractions represent the same number.
- To reduce a fraction, divide numerator and denominator by their greatest common divisor.
- Mixed numbers and improper fractions can be converted back and forth.
- Rational numbers can be written as terminating or repeating decimals.
- To multiply rational numbers, multiply across.
- To divide rational numbers, multiply by the reciprocal.
- To add or subtract rational numbers with unlike denominators, use the LCD.
- Between any two rational numbers, there is another rational number.
- Rational numbers are useful in real-life applications such as scaling recipes.